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1. What Exactly Is Bittensor?

Bittensor is a decentralized, open-source network whose purpose is to produce, rank, and price machine intelligence. It runs on its own blockchain (called Subtensor, built on the Polkadot SDK) and coordinates thousands of independent machines that perform artificial-intelligence work for the network. Unlike a normal AI company, Bittensor does not employ engineers, own data centers, or sell a product. Instead, it maintains a market: anyone who produces useful intelligence gets paid in the network's native token, TAO, in proportion to how valuable that work is judged to be. In the same way that Bitcoin converts electricity into scarce digital money, Bittensor converts compute, models, and machine-learning skill into a tradable digital commodity: intelligence itself.

The project was created by the OpenTensor Foundation, founded by Jacob Steeves and Ala Shaabana, and the network began issuing blocks in early 2021. What started as a single machine-learning market was redesigned in April 2023 with the Finney upgrade, which introduced subnets — independent, self-contained markets that each define their own task and reward system. This turned Bittensor from one application into a whole economy of specialized AI markets sharing one settlement layer. The network's consensus mechanism, Yuma Consensus, is named after the pseudonymous identity used by its founders in early writings, echoing the tradition of Satoshi Nakamoto: the chain itself, not any company, decides who gets paid and how much.

21,000,000	Early 2021	41 / 41 / 18
Hard cap of TAO, mirroring Bitcoin	Network genesis — first TAO issued	Miners / validators / owners: emission split

The mental model matters more than the terminology. Bitcoin takes a physical input (electricity), burns it on an objective competition (hashing), and converts the winner's effort into a scarce digital asset. Bittensor takes the inputs of the AI economy — GPUs, models, data pipelines, and engineering skill — runs them through an objective competition judged by validators, and converts the winners' output into TAO. There is no CEO, no roadmap committee, and no API key to apply for. The network's 'product' is whatever hundreds of competing subnet teams can prove, block after block, to be the most useful intelligence available. That is why supporters describe Bittensor as the first open market for digital intelligence rather than another blockchain project.

> *Bittensor treats intelligence the way Bitcoin treated energy: a raw input that a free market continuously converts into a scarce, tradable asset.*

2. What Does Bittensor Actually Do?

Strip away the branding and Bittensor does three things continuously, every few minutes, forever. First, it lets anyone plug AI workloads into a global market: a miner registers a machine, points it at a subnet, and starts serving whatever that subnet demands — answers, forecasts, embeddings, storage, or raw GPU compute. Second, it automatically scores that work: validators query every miner, compare outputs against the subnet's reward model, and write their judgments to the blockchain as weight vectors. Third, it pays: Yuma Consensus aggregates those judgments into a tamper-resistant ranking, and the chain releases tokens to the ranked participants at every epoch. Produce, rank, pay — that loop is the entire engine.

What does the network literally produce? Today the answer is a long list that grows with every new subnet: large-language-model inference served from decentralized GPU fleets, fine-tuned and pre-trained models, text-prompting services, AI assistants, financial forecasts and trading signals, scraped and structured datasets, vector embeddings, content generation including video, peer-to-peer storage, and general-purpose compute marketplaces. Each subnet is effectively a vertical SaaS product whose suppliers are paid in tokens instead of invoices, and whose customers are the stakers who back the subnet's economy. The practical consequence is that a single network now supplies services that would otherwise require a dozen separate vendor relationships.

Why is that valuable? A centralized AI provider decides internally whether its model is good, prices it however the market tolerates, and can throttle or censor access at will. Bittensor inverts every one of those properties: quality is decided by many independent validators reaching consensus, pricing emerges from open competition for emission, and access is permissionless by construction. Consumers get redundant, commodity-priced intelligence; producers get global demand without sales contracts; and neither side needs anyone's permission. The table below summarizes the contrast.

Dimension	Centralized AI (closed vendor)	Bittensor (open market)
Who decides quality	The vendor's internal evaluation teams	Independent validators reach on-chain consensus
Access	Contracts, sales calls, API keys, possible throttling	Permissionless — register a key and participate
Pricing	Fixed list prices set by the vendor	Market-set emission; capital flows to the best subnets
Source of innovation	Internal R&D; roadmaps	Global open competition across 128+ subnet teams
Payment rail	Fiat invoices and billing cycles	TAO paid per block, within minutes
Failure mode	Outages, censorship, vendor lock-in	Underperformers lose emission or deregister

Table 1 — Structural comparison: closed AI vendors versus the Bittensor market.

3. TAO: The Token That Powers the Network

TAO is the native unit of the Bittensor network, and its name comes from the Tao — the ineffable principle of the I Ching that the founders cite as inspiration for a system that orders itself without a ruler. TAO serves three roles at once: it is the reward miners and validators earn, the stake that validators and delegators lock to secure honest evaluation, and the settlement asset in which every subnet's economy is ultimately denominated. Crucially, the supply was not pre-mined: TAO enters circulation exclusively as emission to network participants who are actively producing or securing intelligence.

The supply design mirrors Bitcoin deliberately. Total supply is hard-capped at exactly 21,000,000 TAO, and issuance follows a decreasing schedule governed by halvings. Unlike Bitcoin, Bittensor's halvings trigger on supply milestones rather than fixed block heights: when the network had issued 10.5 million TAO — half the cap — in December 2025, the block reward was cut in half from 1 TAO to 0.5 TAO, dropping daily issuance from 7,200 TAO to 3,600 TAO. The next milestone is 15.75 million TAO (three quarters of the cap), which at the post-halving rate arrives around 2030, and each subsequent halving slows the faucet further. The final fractions of the 21 million are issued on an asymptotic curve stretching decades into the future — engineered scarcity, exactly as Bitcoin's architects designed it.

21M / ≈11.3M	3,600 TAO	≈70%
Hard cap versus circulating supply	Daily issuance after the Dec 2025 halving	Share of liquid TAO currently staked

So how many TAO are available right now? As of late October 2026, roughly 11.3 to 11.4 million TAO are in circulation — a little over half of the 21 million cap, with the remaining 46 percent scheduled to be emitted over the coming decades. The market values that float at around 2.9 billion US dollars, with TAO trading near 255 dollars per token on the official network interface. Importantly for miners and investors, roughly 70 percent of circulating TAO is staked to validators and subnets, which means the genuinely liquid float is only a few million tokens. Scarcity is therefore double-layered: the issuance schedule caps future supply, and staking removes most of the existing supply from daily trading.

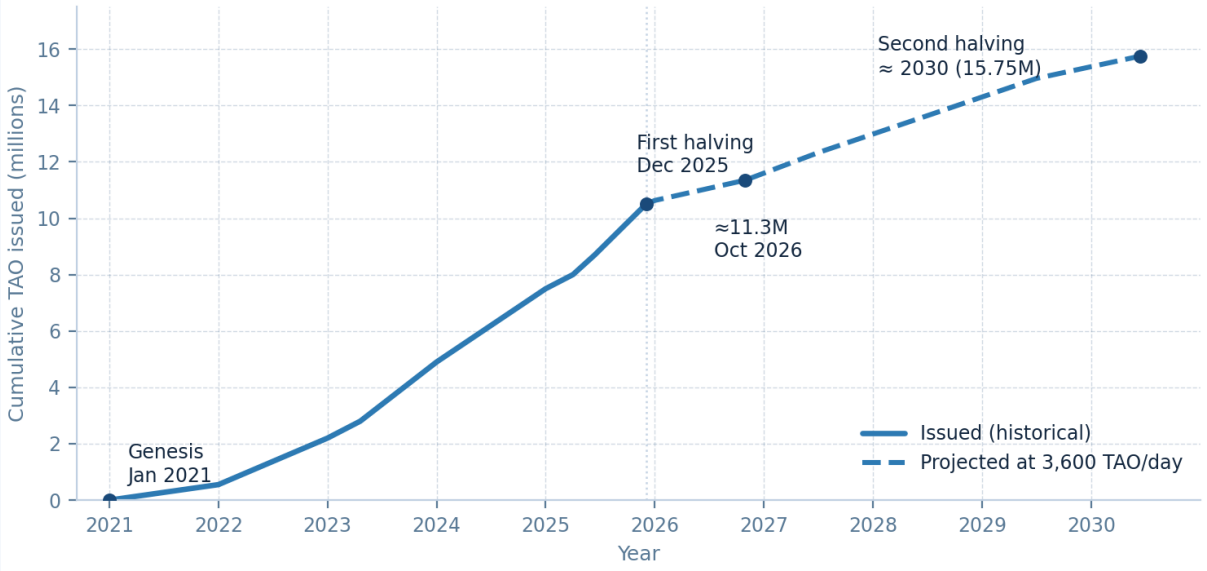


Figure 1 — Cumulative TAO issuance: historical supply to October 2026 and projected path to the second halving at 15.75M (≈2030).

Milestone	Supply trigger	When	Daily issuance after
Genesis	0 TAO issued	January 2021	7,200 TAO per day
First halving	10.5M TAO issued	December 2025	3,600 TAO per day
Second halving	15.75M TAO issued	≈ 2030 (projected)	1,800 TAO per day
Terminal	21M asymptotic limit	Decades away	Approaches zero

Table 2 — TAO supply schedule: halvings trigger on supply milestones, not dates.

4. Subnets: The Markets of Bittensor

A subnet is a self-contained digital market living inside Bittensor, identified by a number called its netuid. Each subnet defines three things: a task (what miners must deliver), a reward model (how quality is measured), and an economy (how emission and stake flow around it). The mainnet is currently capped at 128 active subnet slots, with an expansion to 256 planned — which means a subnet slot is a scarce, contested asset, and the weakest subnets risk being deregistered and recycled when the chain is full. Subnets were introduced by the Finney upgrade in April 2023 and are the reason Bittensor behaves like an economy rather than a single application.

In February 2025 the network activated dynamic TAO (dTAO), the upgrade that gave each subnet its own token economy. Every subnet now issues an alpha token with its own 21 million cap and its own halving curve starting from launch. TAO holders stake into the subnets they believe in; the more capital is staked on a subnet, the larger the share of network emission it captures; and each subnet runs an alpha/TAO liquidity pool that gives its alpha a live market price. The result is a Darwinian capital market layered on top of the

technical market: subnets that attract stakers grow, and subnets that cannot convince capital decay. By March 2026 the combined market capitalization of all subnet alpha tokens had reached roughly 1.12 billion dollars — about 27 percent of TAO's own market cap — and subnets generated an estimated 43 million dollars of economic activity in the first quarter of 2026 alone.

What do these markets actually sell? The variety is the point: the network has become a portfolio of specialized AI businesses. Some subnets serve model inference from distributed GPU fleets; others pre-train or fine-tune models; others run prompting services, AI assistants, trading and forecasting systems, video generation, data pipelines, or storage networks. Measured live inside the Infranex BT platform in October 2026, some of the strongest evidence-backed opportunities included SN82 Noah (video generation), SN5 Hone (assistant services), SN83 CliqueAI, and SN9's pre-training market — each with different hardware requirements, reward mechanics, and competitive dynamics.

Category	What miners deliver	Live examples (Oct 2026)
Inference and serving	LLM completions and API responses from GPU fleets	Compute marketplaces; assistant subnets
Training and pre-training	Model checkpoints, fine-tunes, gradients	SN9 pre-training market
Prompting and assistants	Prompt completions, agent behavior, chat services	SN5 Hone; SN83 CliqueAI
Finance and forecasting	Price predictions, trading signals, risk models	SN68 NOVA; SN107 Minos
Video and media	Generated and processed video content	SN82 Noah
Data, storage and infra	Scraped data, embeddings, storage, raw compute	Data-scraping and storage subnets

Table 3 — Subnet categories and examples of what each market pays miners to produce.

For a miner or investor, two properties of subnets matter most. First, independence: each subnet is a separate labor market with its own demand curve, so a GPU that is mediocre in one market can be elite in another. Second, mortality: because slots are capped, a subnet that loses its stakers and its emission can be removed — the network recycles its weakest markets the way an exchange delists dead stocks. Both properties make subnet selection the single most consequential decision a participant makes, which is precisely the problem Infranex BT's intelligence tooling is built to solve.

5. How a Subnet Works: The Incentive Loop

Every subnet is built from a small set of on-chain primitives. The subnet owner publishes a specification and chain code that define the task and the reward function. Participants run neurons: miners run the worker side, exposing their service through an Axon — a public API endpoint on their machine — while validators run the evaluation side, querying miners with a client called a Dendrite. The entire state of who is participating and how they are scored is visible in the metagraph, the subnet's on-chain directory of neurons, stakes, and weights. Nothing is hidden: a miner can see its rank, its scores, and exactly which validators are rewarding it.

The loop runs on a fixed heartbeat called the tempo — roughly 360 blocks, or about twelve minutes. During each tempo, validators continuously probe miners, score their outputs against the subnet's reward model, and submit weight vectors to the chain. Yuma Consensus then aggregates those vectors into a single authoritative ranking, designed to resist collusion: validators who set weights far from the consensus are penalized, while honest agreement is reinforced through a bond mechanism. Once consensus is reached, the subnet's alpha emission is distributed — by design roughly 41 percent to miners, 41 percent to validators, and 18 percent to the subnet owner. Layered on top, dTAO directs the network's TAO emission into the liquidity pools of the subnets that stakers back, and the chain recycles excess TAO by buying alpha from those pools, supporting each subnet's price floor.

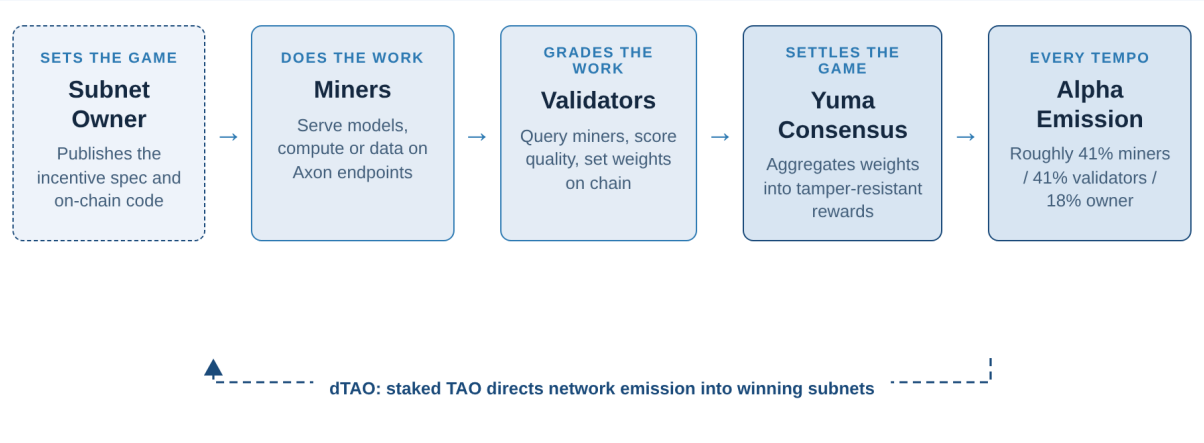


Figure 2 — The subnet incentive loop: owners define, miners serve, validators grade, consensus settles, and dTAO routes capital.

The economic consequences are immediate and unforgiving. Performance is income: a miner's earnings are a direct function of its consensus rank, so quality, latency, and reliability convert into TAO every twelve minutes. Reachability is life: if a miner's Axon endpoint is not publicly reachable, validators cannot score it, and it earns exactly zero no matter how good its hardware is. Competition never sleeps: because weights and emission re-set at every tempo, a miner that rests on yesterday's quality is silently overtaken. This is why professional mining is an operations discipline as much as a machine-learning one — and why the monitoring and verdict tooling described later in this guide exists.

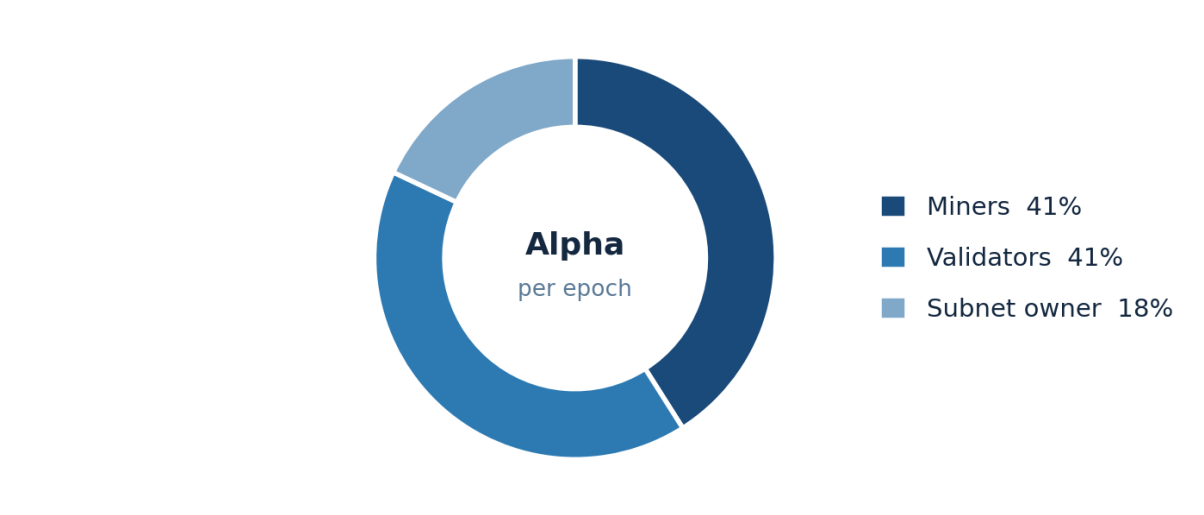


Figure 3 — Intra-subnet alpha emission split: roughly 41% miners, 41% validators, 18% subnet owner each epoch.

6. Validators and Miners: Who Runs the Network

Miners are the network's producers. A miner registers a hotkey into a subnet, runs the subnet's current miner software on its hardware, and serves whatever the subnet demands through its Axon endpoint. Competition is continuous and anonymous: validators do not care who operates a machine, only how its outputs rank against the subnet's reward model on quality, latency, and reliability. Income follows consensus rank — top-ranked miners on a healthy subnet can out-earn the median by an order of magnitude, while bottom-decile miners may earn close to nothing. Entry barriers are modest (a small, variable registration burn plus hardware and skill), which is exactly why performance discipline separates earners from spectators.

Validators are the network's judges — and its capital layer. A validator locks TAO as stake and, once it holds a validator permit (on the order of 1,000 TAO, historically), runs evaluation software that probes every miner in a subnet and submits weight vectors each tempo. Honest, accurate scoring is enforced economically: Yuma Consensus rewards validators whose weights align with consensus and penalizes those who deviate, so a lazy or colluding validator bleeds income. Validators and their stakers share roughly 41 percent of the subnet's emission, distributed by bond weight. Passive TAO holders participate through delegation: they nominate stake to a validator, and returns flow back proportionally — this is also the mechanism through which dTAO lets stakers direct emission toward the subnets they believe in.

The third role, often overlooked, is the subnet owner: the team that designs the incentive mechanism, maintains the code, and earns about 18 percent of the subnet's emission as the cost of building the market. Together the three roles form a system of checks: miners cannot self-declare quality, validators cannot cheaply collude against consensus, and owners cannot move the goalposts arbitrarily without losing the stakers who fund their subnet. The chain itself is the referee — every score, weight, and payout is public in the metagraph, which is what makes the entire economy auditable by tools like Infranex BT.

Role	Entry cost	What they do	Income source	Main risk
Miner	Small registration burn + hardware	Serve the subnet's task on an Axon endpoint	≈41% of subnet emission by rank	Rank collapse; deregistration
Validator	Permit (≈1,000 TAO) + stake	Probe and score miners; set weights	≈41% share, bond-weighted	Penalties for off-consensus scoring
Subnet owner	Development + slot registration	Design and maintain the incentive market	≈18% of emission	Losing stakers; subnet recycled
Staker / delegator	TAO stake only	Back validators and subnets via dTAO	Proportional emission share	Alpha/TAO price swings

Table 4 — The four participation roles: costs, jobs, income, and risks.

7. Why Bittensor Is a Powerful Business Model for Infranex BT

Four structural properties make Bittensor unusually business-friendly for a platform like Infranex BT. First, direct GPU monetization: idle or rented compute can be pointed at a subnet and begin earning within minutes, converting capital expenditure into token revenue without sales cycles. Second, revenue arrives in a hard-capped asset: every payout is TAO or subnet alpha, both governed by Bitcoin-style scarcity, so earnings potentially compound if the asset appreciates. Third, permissionless global scale: there are no contracts, no KYC gates, and no regional restrictions — a machine joins by registering a key. Fourth, and most unusual, the network pays its suppliers first-class: roughly 41 percent of all emission is reserved for miners by protocol design, not by negotiation.

The deeper edge, however, is informational. Bittensor's 128 subnets each have different hardware requirements, reward curves, validator politics, code-update rhythms, and competitive intensities — and public dashboards only scratch that surface. The alpha is in deciding which subnet a given GPU should serve, when to switch, and when to walk away. That is precisely the problem Infranex BT is built around: the Opportunity Engine scores all 128 live subnets on a weighted seven-factor model (net TAO potential, miner competitiveness, reward sustainability, validator acceptance, GPU cost efficiency, operational reliability, and protocol-economic risk), gates every verdict behind hard evidence, and publishes RUN / WATCH / AVOID decisions with reasons attached.

128	7 factors	Daily
Subnet markets, one decision per GPU	Weighted score behind every verdict	Snapshots feeding the trust loop

Measured reality from the platform shows why selection dominates participation. In the October 2026 live snapshot, the engine classified only a handful of the 128 subnets as SCALE-grade opportunities; the strongest evidence-backed picks were SN82 Noah (video generation, projected around 12,700 dollars per month on suitable hardware at the time), SN5 Hone, and SN83 CliqueAI, with a second tier of TEST-grade markets including SN68 NOVA, SN9, and SN107 Minos. The rest of the field was WATCHLIST or explicitly AVOID — negative net earnings, unverified payouts, or hardware-incompatible. A miner that deploys without this filter is rolling dice; a miner that deploys with it is allocating capital.

Beyond mining, the same data opens second-order businesses. dTAO made TAO itself productive, so a treasury strategy — staking into well-chosen subnets and alpha markets — is now a first-class revenue line alongside compute. Subnet ownership offers the 18 percent emission share to teams that can build and defend a market slot. And the platform's daily snapshots and trust loop, which compare projected versus actual earnings for every deployment, compound into a proprietary dataset that competitors cannot buy — the classic data moat, but denominated in on-chain truth.

Honesty requires naming the risks. TAO's price is volatile in both directions; per-subnet emission decays as alpha halvings progress; the 128-slot cap means a weak subnet can be deregistered, stranding miners who depend on it; validator stake remains concentrated enough that reputation matters; and each halving tightens the emission tap, raising the bar for efficiency. None of these are disqualifying — they are the reasons disciplined intelligence, rather than raw participation, is the business.

Infranex BT module	Business function	Value produced
Miner Fit	Match hardware profiles to subnet requirements	No wasted capital on incompatible deployments
Opportunity Engine	Seven-factor scoring, verdicts, and difficulty grading	Capital goes only where evidence supports it
Deployments	One-click provisioning with spend caps and projections	Speed to market with enforced risk limits
Earnings and Wealth	Per-hotkey revenue tracking in TAO and USD	Closed feedback loop between forecast and actual
DevOps engine	Uptime, Axon reachability, self-healing services	The earn-zero failure mode is engineered out
Daily snapshots	Economics history per subnet, trust calibration	A compounding proprietary data moat

Table 5 — How each Infranex BT module converts network data into business advantage.

8. The Miner's Playbook: From GPU to TAO

Everything above reduces to a practical sequence. Getting a GPU to earn in Bittensor is not mysterious — it is a checklist, and every failed miner can trace zero earnings to skipping one of its steps. The playbook below is the same one the Infranex BT platform automates; running it by hand first is the best way to understand what the automation is worth.

1. Match hardware to a subnet before doing anything else. Check VRAM, GPU generation, bandwidth, and whether the subnet demands GPUs at all — the Miner Fit view grades all 128 markets against your exact profile and flags hard incompatibilities before you spend money.

2. Register a hotkey in the chosen subnet. Registration burns a small, variable fee that rises with demand for that subnet; confirm the current fee before committing, and register directly rather than through intermediaries so you hold the keys.

3. Make the Axon endpoint publicly reachable. Open the port, set the firewall, and verify from outside your network. Unreachable Axons are the single most common cause of mining for zero — the chain cannot score what it cannot contact.

4. Run the subnet's current miner code — and keep it current. Subnet teams ship protocol updates that can change models, schemas, or scoring overnight; miners who miss an update fall off the ranking within hours. Pin the release channel and watch for announcements.

5. Track per-hotkey earnings daily and verify payouts on chain, not in a dashboard's promise. Compare realized emission against the projection you deployed on; a gap is the earliest signal that competition or mechanics have shifted.

6. Iterate or switch. If tuned performance stays in the bottom decile after a fair run, move the GPU. The registration burn is small compared to weeks of sub-median emission — the expensive mistake is staying in the wrong market, not moving.

Six factors decide whether a miner climbs the ranking. Reachability and latency come first because they gate everything else; output quality is what the reward model actually scores; uptime compounds trust with validators; version discipline protects accumulated position; and subnet selection — the meta-factor — determines whether excellence is worth anything at all in that market. Infranex BT automates the monitoring of all six, and the roadmap extends this: a registration desk for one-click onboarding, continuous Axon probes that alert before validators notice, per-hotkey earnings reconciliation, deregistration-risk alarms, and subnet code-update alerts.

Success factor	Why it matters	How Infranex BT helps
Axon reachability	Unreachable miners earn zero regardless of quality	Deployment checks plus planned continuous probes
Output quality	The reward model pays exactly what it scores	Subnet verdicts flag competitive intensity first
Latency	Validators time out slow responses under load	Hardware fit scoring prevents overloaded setups
Uptime	Missed tempo windows permanently cost rank	DevOps self-healing and performance-lag alerts
Version updates	Stale miners drop off after protocol changes	Planned subnet update alarms
Subnet selection	Excellence in a bad market still pays nothing	Seven-factor engine with evidence-gated verdicts

Table 6 — What separates earning miners from zero-balance miners.

Costs and risks deserve the same clarity as the upside. Direct costs include the registration burn, GPU capital or rental (A100-class machines rent in the range of a dollar or two per hour on most markets), electricity, and the engineering time to stay current. Structural risks include subnet deregistration, mid-flight protocol changes, emission decay as alpha halvings progress, and TAO price volatility on the revenue line. The playbook's answer to all of these is the same: measure continuously, verify on chain, and pre-commit to switching rules so that exit decisions are made by evidence rather than by hope.

9. Quick Reference and Glossary

The numbers below snapshot the network as of late October 2026, compiled from public network dashboards and live platform telemetry. They are approximate by nature and will drift with the market; the structural facts — the 21 million cap, the 41/41/18 split, the halving schedule — are fixed by protocol and can be relied on. Before any financial decision, verify current values on the official interfaces listed at the end of this guide.

Key number (late Oct 2026)	Value
TAO price / market cap	≈ $255 / ≈ $2.9B
Circulating supply	≈ 11.3M of 21,000,000 hard cap
Staked share of circulating TAO	≈ 70%
Daily TAO emission (post-halving)	3,600 TAO
First / second halving	Dec 2025 (10.5M) / ≈ 2030 (15.75M)
Subnet slots	128 active; expansion to 256 planned
Alpha token market cap (Mar 2026)	≈ $1.12B
Subnet economic activity (Q1 2026)	≈ $43M
Miner share of subnet emission	≈ 41%

Table 7 — Network key numbers at a glance.

Term	Meaning
TAO	Bittensor's native token; 21M hard cap, Bitcoin-style halvings
dTAO	Dynamic TAO — the Feb 2025 upgrade giving subnets token economies
Alpha token	A subnet's own 21M-cap token, priced against TAO in its pool
Subnet	A self-contained market with its own task, rewards, and economy
Netuid	A subnet's numeric ID on the chain (SN82 = netuid 82)
Neuron	Any registered participant (miner or validator) in a subnet
Axon	A miner's public API endpoint that validators query
Dendrite	The validator-side client used to probe miners
Metagraph	On-chain directory of a subnet's neurons, stakes, and weights
Tempo	The epoch heartbeat (≈360 blocks, ≈12 minutes) when weights settle
Yuma Consensus	The collusion-resistant mechanism aggregating validator weights
Hotkey / coldkey	Operational key for mining; offline key holding the stake
Stake	TAO bonded to a validator or subnet; the lever dTAO votes with
Registration	Committing a hotkey to a subnet for a small burn fee
Emission	Newly issued TAO/alpha paid to participants each epoch
Deregistration	Removal of a subnet or neuron when slots are contested

Table 8 — Working glossary of Bittensor terms.

For deeper study, three sources cover almost everything: the official documentation at bittensor.com for protocol mechanics and dTAO details, the network interface at bittensor.ai for live subnet dashboards and token data, and taostats.io for chain analytics. Inside the Infranex BT platform, the Miner Fit and Opportunities views translate this guide's concepts into live, scored decisions on all 128 subnets — which is the point of having built it: the guide explains the machine, and the machine works the market.

9.1 Where to go next

- bittensor.com/docs — protocol documentation: emissions, dTAO, Yuma Consensus, and the official subnet developer guides; start here for mechanism-level detail.
- bittensor.ai — the network's front page: live TAO price, per-subnet dashboards, alpha pools, and registration status for all 128 slots.
- taostats.io — chain analytics: supply history, emission flows, validator distributions, and historical metagraph data for any subnet.
- Infranex BT — the Miner Fit view for hardware matching, the Opportunities view for scored verdicts across all subnets, and Deployments for execution with enforced spend caps; the platform operationalizes every step of the playbook in chapter 8.

A final orientation for the reader: Bittensor rewards people who verify rather than trust, and who treat every twelve-minute epoch as fresh evidence. Use this guide to build the mental model, use the official documentation to verify every mechanism it summarizes, and use the platform to keep the model synchronized with a market that reprices intelligence twice an hour. The network is young, the emission is front-loaded, and the tools are now in place — the remaining variable is disciplined participation.